

[illegible]

Section I

10 marks

Attempt Questions 1 to 10

Allow approximately 15 minutes for this section

Mark your answers on the answer grid provided.

Questions

1. If $w = a + 2ai$, which of the following is equivalent to i^3w ?

- A. $2a - ai$
- B. $a + 2ai$
- C. $a - 2ai$
- D. $2a + ai$

2. The sign on a fence reads:

**"If you are not authorised,
you may not enter."**

Which of the following statements is logically equivalent to the one above?

- A. If you may enter, then you are authorised
- B. If you are authorised, then you may enter
- C. If you are not authorised, then you may enter
- D. If you may not enter, then you are not authorised

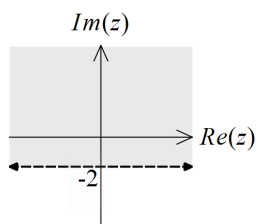
3. Consider the following statement: $\forall x, y \in \mathbb{R}, xy \leq 9 \Rightarrow x \leq 3 \text{ or } y \leq 3$.

Which of the following is the negation of this statement?

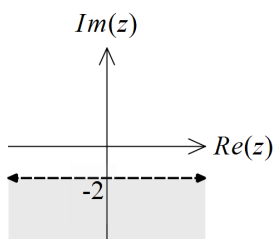
- A. $\forall x, y \in \mathbb{R}, x \leq 3 \text{ or } y \leq 3 \Rightarrow xy \leq 9$
- B. $\forall x, y \in \mathbb{R}, x > 3 \text{ and } y > 3 \Rightarrow xy > 9$
- C. $\exists x, y \in \mathbb{R}, \text{ with } x > 3 \text{ or } y > 3 \text{ such that } xy \leq 9$
- D. $\exists x, y \in \mathbb{R}, \text{ with } x > 3 \text{ and } y > 3 \text{ such that } xy \leq 9$

4. Which of the following is the region in the Argand diagram specified by $iz - i\bar{z} > 4$?

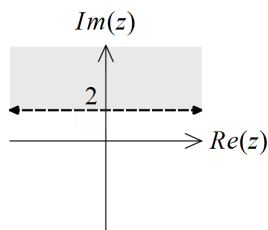
A.



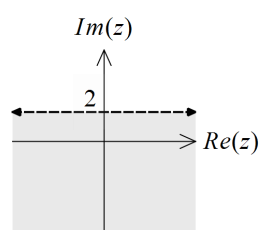
B.



C.



D.

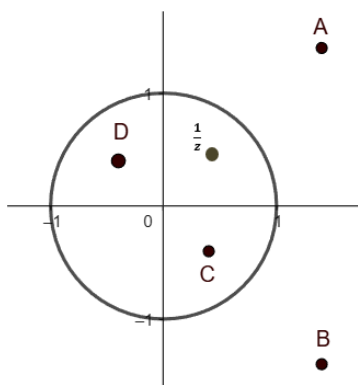


5. A vector makes equal angles of $\frac{\pi}{4}$ with the positive directions of the x -axis and the y -axis. What angle does the vector make with the positive direction of the z -axis?

A. 0

B. $\frac{\pi}{4}$ C. $\frac{\pi}{2}$ D. $\frac{3\pi}{4}$

6. The complex number $\frac{1}{z}$ is shown on the Argand diagram below:



Which one of the points could correspond to the complex number z ?

A. A

B. B

C. C

D. D

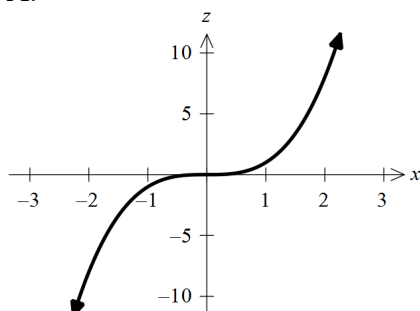
7. Given that $2 - i$ is a root of $P(z) = z^4 - 2z^3 + az^2 + bz + 50$, where $a, b \in \mathbb{R}$, which one of the following is a factor of $P(z)$?

- A. $z^2 + 4z - 5$
- B. $z^2 + 4z + 5$
- C. $z^2 + 2z - 10$
- D. $z^2 + 2z + 10$

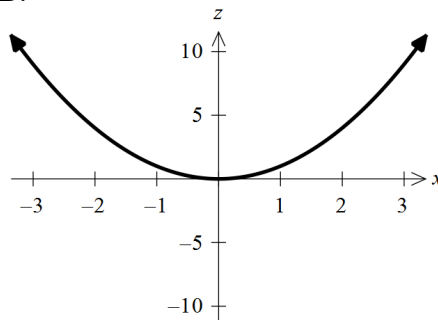
8. The curve $r(t) = (t^2, t^4, t^6)$ is defined for all real values of t .

Which of the following graphs best represents its **projection onto the xz -plane**?

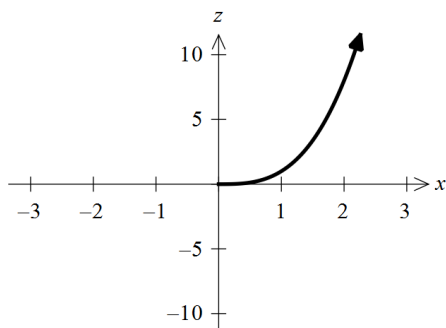
A.



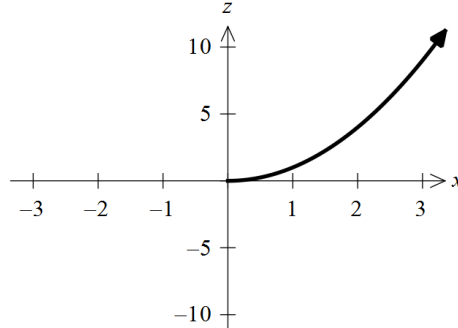
B.



C.



D.



9. Which of the following is equal to the integral below?

$$\int_0^{\frac{\pi}{2}} \frac{1}{1 + \sin \theta + \cos \theta} d\theta$$

- A. $\int_0^1 \frac{1}{1 + \theta} d\theta$
- B. $\int_0^1 \frac{2}{2 + 2\theta - \theta^2} d\theta$
- C. $\int_0^{\frac{\pi}{2}} \frac{1 + \theta^2}{1 + \theta} d\theta$
- D. $\int_0^{\frac{\pi}{2}} \frac{1 - \sin \theta - \cos \theta}{2 - 2\sin \theta \cos \theta} d\theta$

10. The point P with position vector $\overrightarrow{OP} = 4\mathbf{i} + 5\mathbf{j} + 6\mathbf{k}$ lies on a sphere with centre O . Which one of the following points also lies on the sphere?

- A. A with position vector $\overrightarrow{OA} = 2\mathbf{i} + 5\mathbf{j} + 7\mathbf{k}$
- B. B with position vector $\overrightarrow{OB} = 5\mathbf{i} + 5\mathbf{j} + 5\mathbf{k}$
- C. C with position vector $\overrightarrow{OC} = 2\mathbf{i} + 3\mathbf{j} + 8\mathbf{k}$
- D. D with position vector $\overrightarrow{OD} = 2\mathbf{i} + 6\mathbf{j} + 6\mathbf{k}$

End of Section I

Section II

90 marks

Attempt Questions 11 to 16

Allow approximately 2 hours and 45 minutes for this section.

Write your answers in the writing booklets supplied. Additional writing booklets are available. Your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 marks) Start a new writing booklet

Marks

- (a) Consider the complex numbers $w = 2 - i$ and $z = 1 + 3i$.

Find the following in the form $a + bi$, where a and b are real. Show your working.

(i) w^2 1

(ii) $\frac{1}{\overline{w} + z}$ 2

- (b) (i) Find A, B and C such that: 2

$$\frac{8x + 3}{(x - 2)(x^2 + 1)} = \frac{A}{x - 2} + \frac{Bx + C}{x^2 + 1}$$

- (ii) Hence or otherwise, find 2

$$\int \frac{8x + 3}{(x - 2)(x^2 + 1)} dx.$$

- (c) Find the square roots of $12 - 16i$, giving your answers in the form $a + bi$,

where a and b are real numbers. 2

Question 11 continues on the next page.

Question 11(continued)

(d) Consider the vectors $\underline{u} = 2\sqrt{2}\underline{i} + 2\sqrt{2}\underline{j} + 8\underline{k}$ and $\underline{v} = 6\underline{i} + 6\underline{j} + 2\sqrt{2}\underline{k}$

(i) Find the angle between vectors $\underline{u}$ and $\underline{v}$. 2

(ii) Find the projection of the vector $\underline{u}$ onto the vector $\underline{v}$. 2

(e) Find $\int \tan^4 x \, dx$. 2

End of Question 11

Question 12 (15 marks) Start a new writing booklet**Marks**

- (a) Consider the statement: " for $x \in \mathbb{R}$: $x^2 - x > 0 \Rightarrow x < 0$ "
- (i) Show that the statement is false. 1
- (ii) State the converse of the statement. 1
- (iii) Determine, with proof, whether the converse is true. 2
- (b) (i) Express the complex number $-2 + 2i\sqrt{3}$ in complex exponential form with principal argument. 2
- (ii) Hence, or otherwise, find the fourth roots of $-2 + 2i\sqrt{3}$ in simplest complex exponential form with principal arguments. 2
- (c) Show that $\operatorname{Re}\left(\frac{1-z}{1+z}\right) = 0$ for any $z \in \mathbb{C}$ with $|z| = 1$ 2
- (d) An object is moving so that its displacement, x metres, is given by $x = 5\cos^2 t + 4\sin^2 t$ where t is time in seconds.
- (i) Show that the motion is simple harmonic. 2
- (ii) State the centre, amplitude and period of the motion 3

End of Question 12

Question 13 (15 marks) Start a new writing booklet**Marks**

- (a) Consider the line l_1 with cartesian equation $\frac{x-4}{7} = \frac{y+3}{2} = -\frac{z}{6}$

and the line l_2 with parametric vector equation $\vec{r} = \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix} \quad (\lambda \in \mathbb{R})$

- (i) Find the parametric vector equation of l_1 1

- (ii) Show that the lines l_1 and l_2 do not intersect. 2

- (b) Use a proof by contradiction to prove that there are no positive integers p and q

such that $p^2 - q^2 = 2$ 2

- (c) Find $\int \frac{1-3x}{\sqrt{6x-x^2}} dx$ 3

- (d) Sketch the region defined by $|z-i| < |z+i|$ and $-\frac{\pi}{6} \leq \text{Arg}(z-i) \leq \frac{\pi}{6}$ on a **carefully drawn fully labelled** Argand diagram (1/4 page). 3

- (e) Let $I_n = \int_0^\pi x^n \sin x \, dx$.

- (i) Show that $I_n = \pi^n - n(n-1)I_{n-2}$ for $n \geq 2$. 2

- (ii) Evaluate I_4 2

End of Question 13

Question 14 (15 marks) Start a new writing booklet**Marks**

(a) Use the substitution $t = \tan x$ to find $\int \frac{1}{\sin x \cos x} dx$ 3

- (b) A particle moves in simple harmonic motion such that its acceleration is $\ddot{x} = -25(x - 4)$.
Initially the particle is located at $x = 6$ metres from the origin and has a velocity of -10 ms^{-1} .
Determine the range of motion of the particle. 4

- (c) Consider the lines with parametric vector equations:

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} 0 \\ -1 \\ 3 \end{pmatrix} \quad (\lambda \in \mathbb{R}) \text{ and}$$

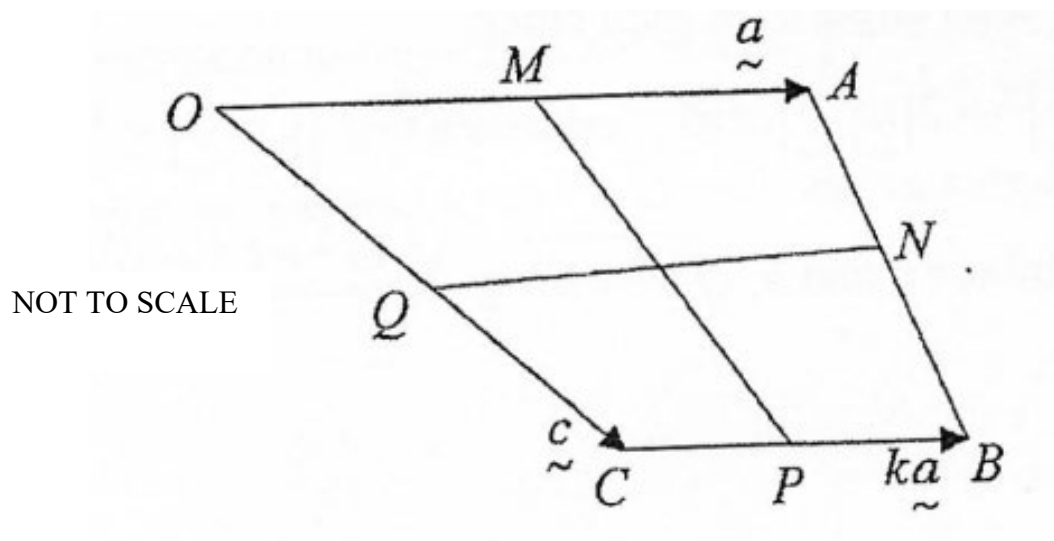
$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix} + \mu \begin{pmatrix} 1 \\ 0 \\ -2 \end{pmatrix} \quad (\mu \in \mathbb{R})$$

Find the point of intersection of the two lines. 3

Question 14 continues on the next page

Question 14 (continued)

(d)



In the diagram $OACB$ is a trapezium in which $\vec{OA} = \vec{a}$, $\vec{OC} = \vec{c}$ and $\vec{CB} = k\vec{OA}$

for some constant $0 < k < 1$.

The points M , N , P , and Q are the midpoints of OA , AB , BC , and CO respectively.

(i) Use vector methods to show that MP and QN bisect each other.

2

(ii) If $MP \perp QN$ use vector methods to show that $|\vec{OC}|^2 = |\vec{AB}|^2$

3**End of Question 14**

Question 15 (15 marks) Start a new writing booklet**Marks**

- (a) Consider the sequence of propositions P_n , $n = 1, 2, 3 \dots$ defined by

$P_n : 14^n$ can be expressed as the sum of the squares of three different positive integers

- (i) Show that P_1 and P_2 are true. 2

- (ii) By considering P_k and P_{k+2} , use Mathematical Induction to prove that P_n is true for all positive integers $n \geq 1$ 3

- (b) (i) Show that $\frac{1+z}{1+z^{-1}} = z$ for $z \neq -1$ 1

- (ii) Hence, find the imaginary part of $\frac{1 + \cos \alpha + i \sin \alpha}{1 + \cos \alpha - i \sin \alpha}$ 2

- (iii) Let $\alpha \in \mathbb{R}$, $n \in \mathbb{Z}^+$ and $B_n = \left(\frac{1 + \sin \alpha + i \cos \alpha}{1 + \sin \alpha - i \cos \alpha} \right)^n$

Show that $B_n = \cos n \left(\frac{\pi}{2} - \alpha \right) + i \sin n \left(\frac{\pi}{2} - \alpha \right)$ 2

- (c) Let $I_n = \int_0^{\frac{\pi}{3}} \tan^{n+1} x \sec x \, dx$. Given that $\int \tan x \sec x \, dx = \sec x$,

- (i) Show that $I_n = \frac{2 \times 3^{\frac{n}{2}}}{n+1} - \frac{n}{n+1} I_{n-2}$ 3

- (ii) Hence find I_2 2

End of Question 15

Question 16 (15 marks) Start a new writing booklet**Marks**

(a) Find $\int \operatorname{cosec}^3 x \sec x \, dx$ **3**

(b) (i) If $x \geq y > 0$, prove that $x^x y^y \geq x^y y^x$. **1**

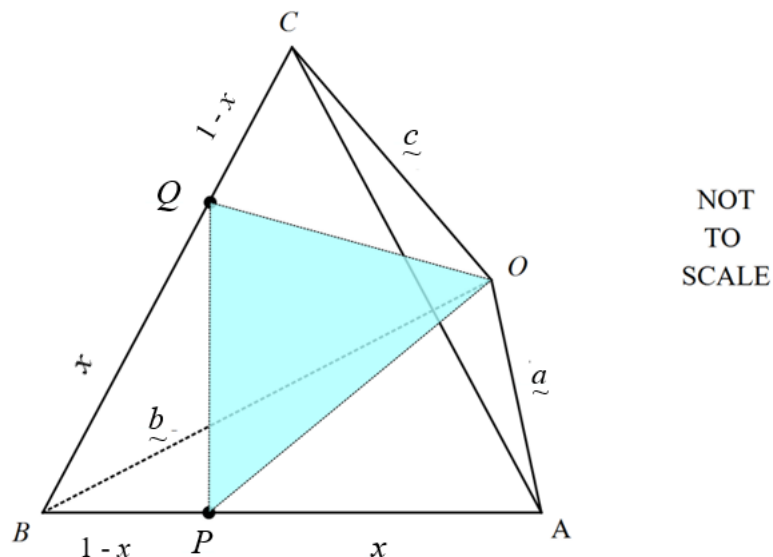
(ii) Hence, or otherwise, prove that for real numbers $a \geq b \geq c > 0$,

$$a^a b^b c^c \geq (abc)^{\frac{a+b+c}{3}}$$
3

Question 16 continues on the next page

Question 16 (continued)

(c) A regular tetrahedron $OABC$ has sides of length 1, where O is the origin.



Let P and Q be the points that divide sides AB and BC in the ratio $x:1-x$ as shown, where $0 < x < 1$.

Let $\overrightarrow{OA} = \underline{a}$, $\overrightarrow{OB} = \underline{b}$ and $\overrightarrow{OC} = \underline{c}$.

It is given that $\underline{a} \cdot \underline{b} = \underline{b} \cdot \underline{c} = \underline{a} \cdot \underline{c} = \frac{1}{2}$

(i) Show that $|\overrightarrow{OP}| = \sqrt{1-x+x^2}$. 2

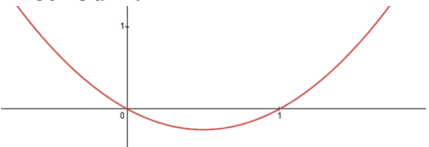
(ii) Let $f(x) = \frac{1+x-x^2}{1-x+x^2}$ where $0 < x < 1$.

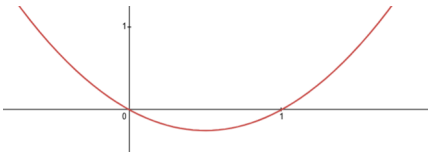
Without using calculus prove that $1 < f(x) \leq \frac{5}{3}$ 3

(iii) Determine the range of values of $\angle POQ$ 3

End of assessment

1	$i^3(a+2ai) = -i(a+2ai) = -ai - 2ai^2 = 2a - ai$	A
2	The contrapositive is: If you may enter, then you are authorised.	A
3	The negation of the given statement must contradict the implication by stating the existence of two real numbers greater than 3 with a product less than or equal to 9	D
4	$i(x+iy) - i(x-iy) = -2y > 4$ $\Rightarrow y < -2$	B
5	If α, β, γ are the direction cosines, then $\cos^2\alpha + \cos^2\beta + \cos^2\gamma = 1$. $\alpha = \beta = \frac{\pi}{4} \Rightarrow \cos^2\gamma = 0 \Rightarrow \gamma = \frac{\pi}{2}$	C
6	Let $\frac{1}{z} = R \operatorname{cis} \theta$, where $0 < R < 1$ and $0 < \theta < \frac{\pi}{2}$. $z = \frac{1}{\frac{1}{z}} = \frac{1}{R \operatorname{cis}(\theta)} = \frac{1}{R} \operatorname{cis}(-\theta)$ $\frac{1}{R} > 1$ and given the argument of $-\theta$ which is a reflection in the x -axis, the point B best represents the complex number z .	B
7	Since the polynomial has real coefficients, any complex roots occur in conjugate pairs. Hence, $2+i$ is also a root. The two roots give the quadratic factor $(z - (2-i))(z - (2+i)) = (z^2 - 4z + 5)$. The other quadratic factor takes the form $(z^2 + pz + q)$, for some real numbers p and q . Since we require the constant term to be 50, the required factor is $z^2 + 2z + 10$.	D
8	$r_{proj_{xz}}(t) = (t^2, 0, t^6) = (x, 0, x^3)$ where $x \geq 0$	C
9	Let $t = \tan \frac{\theta}{2} \Rightarrow d\theta = \frac{2dt}{1+t^2}$ When $\theta = \frac{\pi}{2}, t = 1; \theta = 0, t = 0$ $\therefore \int_0^{\frac{\pi}{2}} \frac{d\theta}{1 + \sin \theta + \cos \theta} = \int_0^1 \frac{1}{1 + \frac{2t}{1+t^2} + \frac{1-t^2}{1+t^2}} \times \frac{2}{1+t^2} dt$ $\int_0^1 \frac{2 dt}{1+t^2+2t+1-t^2} = \int_0^1 \frac{2 dt}{2t+2} = \int_0^1 \frac{dt}{t+1} = \int_0^1 \frac{d\theta}{\theta+1}$	A
10	Radius is $ \vec{OP} = \vec{OC} = \sqrt{77}$ but $ \vec{OA} = \sqrt{78}, \vec{OB} = \sqrt{75}$ and $ \vec{OD} = \sqrt{76}$.	C
11a i	$w^2 = (2-i)^2 = 4 - 4i + i^2 = 3 - 4i$	1 mark
11a ii	$\frac{1}{\overline{w} + z} = \frac{1}{(2+i) + (1+3i)} = \frac{1}{3+4i} = \frac{1}{3+4i} \times \frac{3-4i}{3-4i} = \frac{3-4i}{25} = \frac{3}{25} - \frac{4}{25}i$	2 marks – correct solution 1 mark – attempts to realise denominator
11b i	$8x + 3 \equiv A(x^2 + 1) + (Bx + C)(x - 2)$ Let $x = 2$, then $A = \frac{19}{5}$ Let $x = 0$, then $3 = \frac{19}{5} - 2C \Rightarrow C = \frac{2}{5}$ Let $x = 1$, then $11 = 2\left(\frac{19}{5}\right) - B - \frac{2}{5} \Rightarrow B = -\frac{19}{5}$	2 marks correct solution 1 mark for finding A, B or C

11b ii	$\int \frac{8x+3}{(x-2)(x^2+1)} dx = \frac{19}{5} \int \frac{1}{x-2} dx + \int \frac{-\frac{19}{5}x + \frac{2}{5}}{x^2+1} dx$ $= \frac{19}{5} \ln x-2 - \frac{19}{5} \int \frac{x}{x^2+1} dx + \frac{2}{5} \int \frac{1}{x^2+1} dx$ $= \frac{19}{5} \ln x-2 - \frac{19}{10} \ln(x^2+1) + \frac{2}{5} \tan^{-1} x + C$	2 marks correct solution 1 mark for integrating at least one term correctly
11 c	Let $z = x + iy$, where $x, y \in \mathbb{R}$. Hence, solve the equation $z^2 = 12 - 16i$. $(x + iy)^2 = 12 - 16i$ $x^2 - y^2 + 2xyi = 12 - 16i$ Equating real and imaginary parts, $x^2 - y^2 = 12$ and $xy = -8$ Solving equations simultaneously, $x^2 - \left(\frac{-8}{x}\right)^2 = 12$ $x^4 - 12x^2 - 64 = 0$ Using the quadratic formula, $x = \pm 4, \therefore y = \mp 2$. Hence, the square roots are $4 - 2i, -4 + 2i$.	2 marks correct solution 1 mark for correct equation in x or y
11d i	$\cos\theta = \frac{\tilde{u} \cdot \tilde{v}}{ \tilde{u} \tilde{v} } = \frac{(12+12+16)\sqrt{2}}{\sqrt{80} \times \sqrt{80}} = \frac{1}{\sqrt{2}} \quad \therefore \theta = \frac{\pi}{4}$	2 marks 1 mark for correct numerator or denominator
11 d ii	$\text{proj}_{\tilde{v}} \tilde{u} = \tilde{u} \cos\theta \hat{\tilde{v}} = \sqrt{80} \times \frac{1}{\sqrt{2}} \times \frac{1}{\sqrt{80}} \tilde{v} = 3\sqrt{2} \tilde{i} + 3\sqrt{2} \tilde{j} + 2\tilde{k}$	2 marks 1 mark for partially correct expression for the projection
11e	$\int \tan^4 x dx = \int \tan^2 x (\sec^2 x - 1) dx$ $= \int \tan^2 x \sec^2 x - (\sec^2 x - 1) dx$ $= \frac{1}{3} \tan^3 x - \tan x + x + c$	2 marks 1 mark for correctly obtaining one of the terms of the primitive function
12a i	Method 1: A suitable counterexample is $x = 2$ since $2^2 - 2 > 0$ without requiring a negative value of x. Method 2: 	1 mark

	$x^2 - x > 0$ $x(x - 1) > 0$ $x < 0 \text{ or } x > 1 \text{ (from graph)}$ $\therefore x < 0 \text{ may not be the case}$	
12 a ii	Converse is: "If $x < 0$, then $x^2 - x > 0$ ".	1 mark correct solution
12 a iii	Method 1: Let $x < 0$. Then $x^2 > 0$ and $-x > 0$. Adding these gives $x^2 + (-x) > 0 \rightarrow x^2 - x > 0$ Method 2: Let $x < 0$. Then $x - 1 < 0$. Multiplying these gives $x(x - 1) > 0 \rightarrow x^2 - x > 0$ Method 3: Consider the graph of $y = x^2 - x$.  We can see that for $x < 0$, $x^2 - x > 0$.	2 marks – correct solution 1 mark – a correct method with unclear reasoning
12 b i	$-2 + 2i\sqrt{3} = 4e^{i\frac{2\pi}{3}}$	2 marks – correct 1 mark – correct modulus or argument
12 b ii	Let $z = re^{i\theta}$ be a fourth root. Thus, $z^4 = 4e^{i\frac{2\pi}{3}}$ $z = \sqrt[4]{4}e^{i\frac{(\frac{2\pi}{3} + 2k\pi)}{4}}$ $z = \sqrt{2}e^{i\frac{2\pi(1+3k)}{12}}, k = 0, 1, 2, 3$ That is, $z = \sqrt{2}e^{i\frac{\pi}{6}}, \sqrt{2}e^{i\frac{2\pi}{3}}, \sqrt{2}e^{-i\frac{\pi}{3}}, \sqrt{2}e^{-i\frac{5\pi}{6}}$	2 marks – correct solution 1 mark – finds one correct root

12 c	Let $z = x + iy$ $ \begin{aligned} \operatorname{Re}\left(\frac{1-z}{1+z}\right) &= \operatorname{Re}\left(\frac{1-(x+iy)}{1+(x+iy)}\right) \\ &= \operatorname{Re}\left(\frac{(1-x)-iy}{(1+x)+iy}\right) \\ &= \operatorname{Re}\left(\frac{(1-x)-iy}{(1+x)+iy} \times \frac{(1+x)-iy}{(1+x)-iy}\right) \\ &= \operatorname{Re}\left(\frac{1-x^2-y^2}{(1+x)^2+y^2}\right) \\ &= \operatorname{Re}\left(\frac{1-(x^2+y^2)}{(1+x)^2+y^2}\right) \\ &= \operatorname{Re}\left(\frac{1- z ^2}{(1+x)^2+y^2}\right) \\ &= \operatorname{Re}\left(\frac{1-1}{(1+x)^2+y^2}\right) \text{ as } z =1 \\ &= 0 \end{aligned} $ Method 2: A complex number w is purely imaginary if $w + \overline{w} = 0$ $ \begin{aligned} &\frac{1-z}{1+z} + \overline{\left(\frac{1-z}{1+z}\right)} \\ &= \frac{1-z}{1+z} + \frac{1-\overline{z}}{1+\overline{z}} \\ &= \frac{1+\overline{z}-z-z\overline{z}+1-\overline{z}+z-z\overline{z}}{1+z+\overline{z}+z\overline{z}} \\ &= \frac{2-2 z ^2}{1+2\operatorname{Re}(z)+ z ^2} \\ &= \frac{2-2}{2+2\operatorname{Re}(z)} \\ &= 0 \end{aligned} $ $\therefore \frac{1+z}{1-z}$ is purely imaginary Method 3: Let	2 marks 1 mark for rewriting $\frac{1-z}{1+z}$ using conjugate properties
12 d i	$ \begin{aligned} x &= 5\cos^2 t + 4\sin^2 t \\ x &= 4\cos^2 t + 4\sin^2 t + \cos^2 t \\ x &= 4 + \cos^2 t \\ x &= 4 + \frac{\cos 2t + 1}{2} \\ x &= \frac{9}{2} + \frac{\cos 2t}{2} \\ \dot{x} &= -\sin 2t \\ \ddot{x} &= -2\cos 2t \\ \ddot{x} &= -4\left(x - \frac{9}{2}\right) \end{aligned} $ Since acceleration is proportional to displacement, the motion is simple harmonic.	2 marks 1 mark for finding a correct expression for the acceleration

12 d ii	Centre = 4.5, Amplitude = $\frac{1}{2}$ Period = π	3 marks [1 mark each]
13 a i	$\frac{x-4}{7} = \frac{y+3}{2} = -\frac{z}{6} = \mu$ $x = 4 + 7\mu$ $y = -3 + 2\mu$ $z = -6\mu$ $l_1 = \begin{pmatrix} 4 \\ -3 \\ 0 \end{pmatrix} + \mu \begin{pmatrix} 7 \\ 2 \\ -6 \end{pmatrix}$	
13 a ii	If they meet it would be where (x,y,z) equal on both lines. equating x, y, z components gives $4 + 7\mu = 1 + 2\lambda \quad (1)$ $-3 + 2\mu = \lambda \quad (2)$ $-6\mu = -1 + \lambda \quad (3)$ sub (2) into (3) $-6\mu = -1 - 3 + 2\mu$ $-8\mu = -4$ $\mu = \frac{1}{2}$ $\lambda = -2$ sub into (1) $4 + \frac{7}{2} = 1 - 4$ $7.5 = -3 \text{ not true } \therefore \text{ lines do not intersect}$	
13 b	Suppose that there exist positive integers p and q such that $p^2 - q^2 = 2 \Rightarrow (p - q)(p + q) = 2$ Since p and q are positive integers, $p - q < p + q$ are also integers. Now consider all pairs of positive integers whose product is 2: the only possibility is $p - q = 1$ and $p + q = 2 \Rightarrow p = \frac{3}{2}, q = \frac{1}{2}$ which contradicts p and q being integers. Hence the assumption is wrong and no such positive integers p and q exist.	2 marks - Derives contradiction clearly (e.g., $p = 3/2$) and concludes correctly 1 mark – sets up contradiction and shows $(p-q)(p+q)=2$ with correct factor pair up
13 c	Completing the square in the denominator, $\int \frac{1-3x}{\sqrt{9-(x-3)^2}} dx$	

Let $u = x - 3$,
 $du = dx$

$$\int \frac{1 - 3(u + 3)}{\sqrt{9 - u^2}} du = \int \frac{-3u - 8}{\sqrt{9 - u^2}} du = -3 \int \frac{u}{\sqrt{9 - u^2}} du - 8 \int \frac{du}{\sqrt{9 - u^2}}$$

$$= 3\sqrt{9 - u^2} - 8 \sin^{-1} \frac{u}{3} + C$$

$$= 3\sqrt{9 - (x - 3)^2} - 8 \sin^{-1} \frac{x - 3}{3} + C$$

$$= 3\sqrt{6x - x^2} - 8 \sin^{-1} \frac{x - 3}{3} + C$$

13
d

let $z = x + iy$

$$|z - i| < |z + i|$$

$$|x + iy - i| < |x + iy + i|$$

$$|x + i(y - 1)| < |x + i(y + 1)|$$

$$x^2 + (y - 1)^2 < x^2 + (y + 1)^2$$

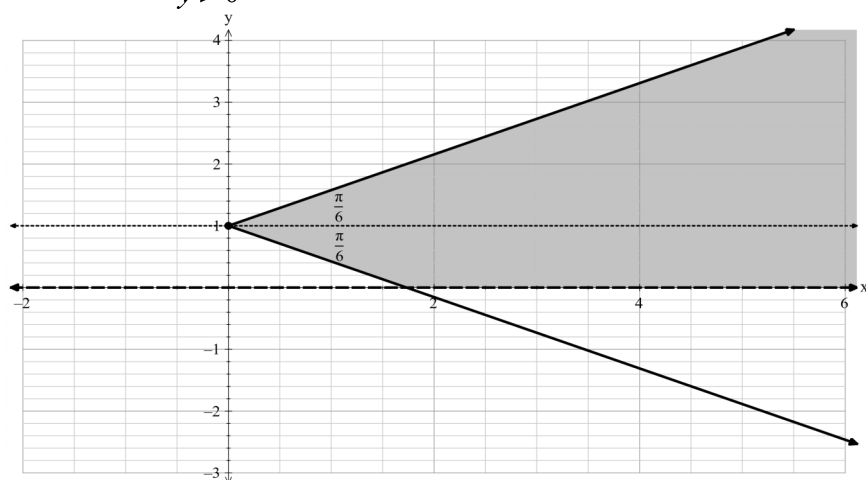
$$x^2 + y^2 - 2y + 1 < x^2 + y^2 + 2y + 1$$

$$-2y < 2y$$

$$0 < 4y$$

$$y > 0$$

(the perpendicular bisector of the interval from $-i$ to i)



13 e i	$I_n = \int_0^{\pi} x^n \sin x \, dx$ $u = x^n \quad dv = \sin x \, dx$ $du = nx^{n-1} \quad v = -\cos x$ $I_n = [-x^n \cos x](\pi, 0) - \int_0^{\pi} -\cos x \, nx^{n-1} \, dx$ $= [-\pi^n \times -1 - 0] + n \int_0^{\pi} x^{n-1} \cos x \, dx$ $[\pi^n] + n \int_0^{\pi} x^{n-1} \cos x \, dx$ $u = x^{n-1} \quad dv = \cos x \, dx$ $du = (n-1)x^{n-2} \, dx \quad v = \sin x$ $I_n = \pi^n + n \left([x^{n-1} \sin x](\pi, 0) - \int_0^{\pi} \sin x \, (n-1)x^{n-2} \, dx \right)$ $I_n = \pi^n + n \left([x^{n-1} \sin x](\pi, 0) - (n-1) \int_0^{\pi} x^{n-2} \sin x \, (n-1) \, dx \right)$ $I_n = \pi^n + n([0 - 0] - (n-1)I_{n-2})$ $I_n = \pi^n - n(n-1)I_{n-2}$	
13 e ii	$I_4 = \pi^4 - 4 \times 3 \times I_2 = \pi^4 - 12I_2$ $I_2 = \pi^2 - 2 \times 1 \times I_0 = \pi^2 - 2I_0$ $I_0 = \int_0^{\pi} x^0 \sin x \, dx = \int_0^{\pi} \sin x \, dx = -[\cos x](\pi, 0) = -(-1 - 1) = 2$ $I_2 = \pi^2 - 4$ $I_4 = \pi^4 - 12(\pi^2 - 4) = \pi^4 - 12\pi^2 + 48$	

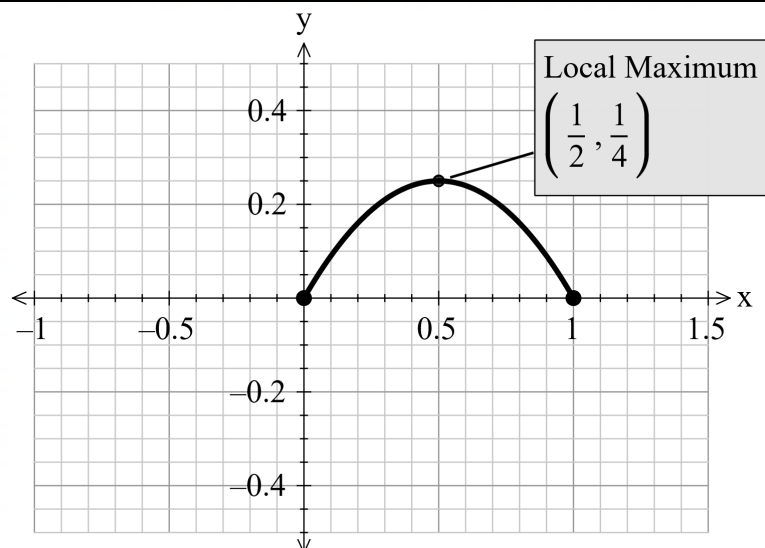
14a	$I = \int \frac{1}{\sin x \cos x} dx = \int \frac{1}{\frac{1}{2}(\sin 2x)} dx$ $= \int \frac{2}{\sin 2x} dx$ let $t = \tan x$ then $\sin 2x = \frac{2t}{1+t^2}$ and $\operatorname{cosec} 2x = \frac{1+t^2}{2t}$ $x = \tan^{-1} t$ $dx = \frac{1}{1+t^2} dt$ now $I = \int \frac{2(1+t^2)}{2t} \frac{1}{1+t^2} dt$ $= \int \frac{1}{t} dt$ $= \ln t + c$ $= \ln \tan x + c$											
14b	<table><tr><th>Criteria</th><th>Marks</th></tr><tr><td>• Correctly solves the inequality and determines the range of motion</td><td>4</td></tr><tr><td>• Substitutes into the velocity equation and sets $v^2 \geq 0$ to find the limits of motion</td><td>3</td></tr><tr><td>• Integrates both sides and uses the initial conditions to find the constant of integration</td><td>2</td></tr><tr><td>• Correctly uses the chain rule to relate acceleration to velocity: $a = v \frac{dv}{dx}$ and sets up the differential equation</td><td>1</td></tr></table> Sample answer: $v \frac{dv}{dx} = -25(x-4) \Rightarrow \frac{1}{2} v^2 = -\frac{25}{2} x^2 + 100x + c$ $x = 6, v = -10 \Rightarrow 50 = -450 + 600 + c$ $\Rightarrow c = -100$ $v^2 = -25x^2 + 200x - 200 = -25(x^2 - 8x + 16) + 200 = 200 - 25(x-4)^2$ $v^2 \geq 0 \Rightarrow 25(x-4)^2 \leq 200$ $\Rightarrow (x-4)^2 \leq 8$ $\Rightarrow -2\sqrt{2} \leq x-4 \leq 2\sqrt{2}$ $\Rightarrow 4-2\sqrt{2} \leq x \leq 4+2\sqrt{2}$ i.e. the particle moves between $x = 4-2\sqrt{2}$ and $x = 4+2\sqrt{2}$	Criteria	Marks	• Correctly solves the inequality and determines the range of motion	4	• Substitutes into the velocity equation and sets $v^2 \geq 0$ to find the limits of motion	3	• Integrates both sides and uses the initial conditions to find the constant of integration	2	• Correctly uses the chain rule to relate acceleration to velocity: $a = v \frac{dv}{dx}$ and sets up the differential equation	1	
Criteria	Marks											
• Correctly solves the inequality and determines the range of motion	4											
• Substitutes into the velocity equation and sets $v^2 \geq 0$ to find the limits of motion	3											
• Integrates both sides and uses the initial conditions to find the constant of integration	2											
• Correctly uses the chain rule to relate acceleration to velocity: $a = v \frac{dv}{dx}$ and sets up the differential equation	1											

14c	$x : 1 = \mu$ ① $y : -\lambda = 1$ ② $z : 3\lambda = -1 + 2\lambda$ ③ $\mu = 1$ $\lambda = -1$ point of intersection is (1, 1, -3)									
14d i	Answer N is the midpoint of AB . $\therefore \vec{ON} = \frac{1}{2}(\vec{OA} + \vec{OB})$ $\therefore \vec{ON} = \frac{1}{2}(\vec{a} + (\vec{c} + k\vec{a})) = \frac{1}{2}\vec{c} + \frac{1}{2}(k+1)\vec{a}$ $\therefore \frac{1}{2}(\vec{OQ} + \vec{ON}) = \frac{1}{2}\left(\frac{1}{2}\vec{c} + \frac{1}{2}\vec{c} + \frac{1}{2}(k+1)\vec{a}\right)$ $= \frac{1}{2}\vec{c} + \frac{1}{4}(k+1)\vec{a}$ But $\frac{1}{2}(\vec{OM} + \vec{OP}) = \frac{1}{2}\left(\frac{1}{2}\vec{a} + \left(\vec{c} + \frac{1}{2}k\vec{a}\right)\right) = \frac{1}{2}\vec{c} + \frac{1}{4}(k+1)\vec{a}$ Hence QN and MP have a common midpoint T where $\vec{OT} = \frac{1}{2}\vec{c} + \frac{1}{4}(k+1)\vec{a}$. Hence QN and MP bisect each other in their intersection point T .									
14 d ii	Answer $\vec{MP} = \vec{MO} + \vec{OC} + \vec{CP} = -\frac{1}{2}\vec{a} + \vec{c} + \frac{1}{2}k\vec{a} = \vec{c} + \frac{1}{2}(k-1)\vec{a}$ $\vec{QN} = \vec{QO} + \vec{ON} = -\frac{1}{2}\vec{c} + \frac{1}{2}\vec{c} + \frac{1}{2}(k+1)\vec{a} = \frac{1}{2}(k+1)\vec{a}$ (using the expression for $\vec{ON}$ found in (i)) Hence if $MP \perp QN$, $\left(\vec{c} + \frac{1}{2}(k-1)\vec{a}\right) \cdot \left(\frac{1}{2}(k+1)\vec{a}\right) = 0$, giving $\vec{c} \cdot \vec{a} = -\frac{1}{2}(k-1)\vec{a} \cdot \vec{a}$ $\vec{AB} = \vec{AO} + \vec{OC} + \vec{CB} = -\vec{a} + \vec{c} + k\vec{a} = \vec{c} + (k-1)\vec{a}$ $\therefore \vec{AB} ^2 = (\vec{c} + (k-1)\vec{a}) \cdot (\vec{c} + (k-1)\vec{a})$ $= \vec{c} \cdot \vec{c} + (k-1)^2 \vec{a} \cdot \vec{a} + 2(k-1)\vec{a} \cdot \vec{c}$ $= \vec{c} \cdot \vec{c} + \left\{(k-1)^2 + 2(k-1)\left(-\frac{1}{2}(k-1)\right)\right\} \vec{a} \cdot \vec{a}$ if $MP \perp QN$ $= \vec{c} \cdot \vec{c}$ $= \vec{OC} ^2$									
15 a i	$14^1 = 1^2 + 2^2 + 3^2$ hence P_1 is true. $14^2 = 12^2 + 6^2 + 4^2$ hence P_2 is true.	1 mark for each P_1 and P_2								
15 a ii	<table border="1"><thead><tr><th>Criteria</th><th>Marks</th></tr></thead><tbody><tr><td>Uses mathematical induction to establish the truth of all the propositions P_n for integers $n \geq 1$</td><td>3</td></tr><tr><td>Substantial progress eg. shows the truth of P_k implies the truth of P_{k+2}</td><td>2</td></tr><tr><td>Some progress eg. attempts to incorporate truth of P_k into consideration of P_{k+1} or P_{k+2}</td><td>1</td></tr></tbody></table> Answer If P_k is true: $14^k = a^2 + b^2 + c^2$ for different positive integers a, b, c . * Consider P_{k+2} : $14^{k+2} = 14^2 \times 14^k$ $= 14^2(a^2 + b^2 + c^2)$ if P_k is true, using * $= (14a)^2 + (14b)^2 + (14c)^2$, where $14a, 14b, 14c$ are different positive integers. Hence, if P_k is true, then P_{k+2} is true. But P_1 is true, therefore by mathematical induction, P_n is true for all odd integers $n \geq 1$. Also P_2 is true, and therefore by mathematical induction, P_n is true for all even integers $n \geq 2$. Hence P_n is true for all integers $n \geq 1$.	Criteria	Marks	Uses mathematical induction to establish the truth of all the propositions P_n for integers $n \geq 1$	3	Substantial progress eg. shows the truth of P_k implies the truth of P_{k+2}	2	Some progress eg. attempts to incorporate truth of P_k into consideration of P_{k+1} or P_{k+2}	1	
Criteria	Marks									
Uses mathematical induction to establish the truth of all the propositions P_n for integers $n \geq 1$	3									
Substantial progress eg. shows the truth of P_k implies the truth of P_{k+2}	2									
Some progress eg. attempts to incorporate truth of P_k into consideration of P_{k+1} or P_{k+2}	1									
15 b i	$\frac{1+z}{1+\frac{1}{z}} = \frac{1+z}{\frac{z+1}{z}} = z \frac{1+z}{1+z} = z$	1 mark – correct solution								

15 b ii	let $z = \cos \alpha + i \sin \alpha$ then $z^{-1} = \cos(-\alpha) + i \sin(-\alpha)$ $= \cos \alpha - i \sin \alpha$ and $\frac{1 + (\cos \alpha + i \sin \alpha)}{1 - (\cos \alpha - i \sin \alpha)}$ $= \frac{1 + z}{1 + z^{-1}} = z = \cos \alpha + i \sin \alpha$ the imaginary part is $\sin \alpha$	
15 b iii	let $z = \sin \alpha + i \cos \alpha$ $B_n = \left(\frac{1 + z}{1 + z^{-1}} \right)^n$ $= z^n$ (from (i)) $= (\sin \alpha + i \cos \alpha)^n$ $= \left(\cos \left(\frac{\pi}{2} - \alpha \right) + i \sin \left(\frac{\pi}{2} - \alpha \right) \right)^n$ (complementary angles) $= \cos n \left(\frac{\pi}{2} - \alpha \right) + i \sin n \left(\frac{\pi}{2} - \alpha \right)$	2 marks – correct solution 1 mark – finds B_n in cis form
15 c i	For $I_n = \int_0^{\frac{\pi}{3}} \tan^{n+1} x \sec x \, dx$ Rewrite as $I_n = \int_0^{\frac{\pi}{3}} \tan^n x \tan x \sec x \, dx$ Using integration by parts, $u = \tan^n x, v' = \tan x \sec x$ $u' = n \tan^{n-1} x \sec^2 x, v = \sec x$ $\therefore \int_0^{\frac{\pi}{3}} \tan^n x \tan x \sec x \, dx = [\tan^n x \sec x]_0^{\frac{\pi}{3}} - n \int_0^{\frac{\pi}{3}} \tan^{n-1} x \sec^3 x \, dx$ $= \left[2 \times 3^{\frac{n}{2}} \right] - n \int_0^{\frac{\pi}{3}} \tan^{n-1} x \sec^2 x \sec x \, dx$ $= \left[2 \times 3^{\frac{n}{2}} \right] - n \int_0^{\frac{\pi}{3}} \tan^{n-1} x (\tan^2 x + 1) \sec x \, dx$ $= \left[2 \times 3^{\frac{n}{2}} \right] - n \int_0^{\frac{\pi}{3}} \tan^{n+1} x \sec x + \tan^{n-1} x \sec x \, dx$ $I_n = \left[2 \times 3^{\frac{n}{2}} \right] - n I_n - n I_{n-2}$ $n I_n + I_n = \left[2 \times 3^{\frac{n}{2}} \right] - n I_{n-2}$ $I_n = \frac{2 \times 3^{\frac{n}{2}}}{n+1} - \frac{n}{n+1} I_{n-2}$	3 marks – correct solution 2 marks – substantial progress – correct integration. 1 mark – uses integration by parts and makes some progress
15 c ii	$I_2 = \frac{2 \times 3^{\frac{2}{2}}}{2+1} - \frac{2}{2+1} I_0 = \frac{6}{3} - \frac{2}{3} I_0 = 2 - \frac{2}{3} I_0$	2 marks – correct solution.

	$I_0 = \int_0^{\frac{\pi}{3}} \tan x \sec x \, dx = [\sec x]_0^{\frac{\pi}{3}} = 1$ $\therefore I_2 = 2 - \frac{2}{3} \times 1 = \frac{4}{3}$	1 mark – finds I_0 or equivalent merit.
16a	$I = \int \operatorname{cosec}^3 x \sec x \, dx$ $= \int \frac{\operatorname{cosec}^2 x}{\sin x \cos x} \, dx$ $= \int \frac{1 + \cot^2 x}{\sin x \cos x} \, dx$ $= \int \frac{1}{\sin x \cos x} + \int \frac{\cot^2 x}{\sin x \cos x} \times \frac{\cos x}{\cos x} \, dx$ $= \int 2 \operatorname{cosec} 2x \, dx + \int \sec^2 x (\tan x)^{-3} \, dx$ $\text{let } I_i = 2 \int \operatorname{cosec} 2x \, dx$ $\text{let } t = \tan x$ $dx = \frac{1}{1+t^2} \, dt$ $\operatorname{cosec} 2x = \frac{1+t^2}{2t}$ $I_i = \int \frac{1+t^2}{2t} \frac{2}{1+t^2} \, dt = \ln t + \underline{c} = \ln \tan x + \underline{c}$ $\text{and } I = \ln \tan x - \frac{1}{2\tan^2 x} + \underline{c}$	3 marks – correct solution 2 marks – substantial progress – partially correct integration 1 mark – some progress. Manipulates integrand so as to make progress towards solution.
16 b i	$x \geq y > 0$ $\therefore \frac{x}{y} \geq 1$ $\text{and } \left(\frac{x}{y}\right)^x \geq \left(\frac{x}{y}\right)^y$ $\frac{x^x}{y^x} \geq \frac{x^y}{y^y}$ $x^x y^y \geq x^y y^x$	1 mark – correct solution
16 b ii	From $x^x y^y \geq x^y y^x$, $a^a b^b \geq a^b b^a$ $b^b c^c \geq b^c c^b$ $c^c a^a \geq c^a a^c$ Multiplying these inequalities, $(a^a b^b c^c)^2 \geq a^{b+c} b^{a+c} c^{a+b}$	3 marks – correct solution 2 marks – multiplies the 3 inequalities correctly 1 mark – uses part i to obtain a suitable inequality

	Multiplying by $a^a b^b c^c$, $(a^a b^b c^c)^3 \geq a^{a+b+c} b^{a+b+c} c^{a+b+c}$ $(a^a b^b c^c)^3 \geq (abc)^{a+b+c}$ Therefore, $a^a b^b c^c \geq (abc)^{\frac{a+b+c}{3}} \text{ as required.}$	
16c i	$\overrightarrow{OP} = \underline{a} + x\overrightarrow{AB}$ $\overrightarrow{AB} = -\underline{a} + \underline{b}$ so $\overrightarrow{OP} = \underline{a} + x(-\underline{a} + \underline{b})$ $= (1-x)\underline{a} + x\underline{b}$ $ \overrightarrow{OP} = \sqrt{\overrightarrow{OP} \cdot \overrightarrow{OP}}$ $\overrightarrow{OP} \cdot \overrightarrow{OP} = ((1-x)\underline{a} + x\underline{b}) \cdot ((1-x)\underline{a} + x\underline{b})$ $= (1-x)^2 \underline{a} \cdot \underline{a} + 2x(1-x)\underline{a} \cdot \underline{b} + x^2 \underline{b} \cdot \underline{b}$ $= (1-x)^2 + x(1-x) + x^2 \text{ (as } \underline{b} \cdot \underline{b} = \underline{b} ^2 = 1 \text{ (same for a) and } \underline{a} \cdot \underline{b} = \frac{1}{2} \text{ (given))}$ $= 1 - 2x + x^2 + x - x^2 + x^2$ $= 1 - x + x^2$ and $\overrightarrow{OP} = \sqrt{1 - x + x^2}$	2 marks – correct solution 1 mark – correct vector expression for OP.
16c ii	$f(x) = \frac{1+x-x^2}{1-x+x^2} = \frac{1+x(1-x)}{1-x(1-x)}$ $= \frac{1+x(1-x)}{1-x(1-x)} \times \frac{1-x(1-x)}{1-x(1-x)}$ $= \frac{1^2 - (x(1-x))^2}{(1-x(1-x))^2}$ Consider $x(1-x)$ for $0 \leq x \leq 1$	3 marks – correct solution 2 marks – progress towards second inequality 1 mark – correctly shows one side of the inequality



In the domain $0 < x < 1$

$$0 < x(1-x) \leq \frac{1}{4}$$

when $x(1-x) = 0$

$$f(x) = \frac{1-0^2}{(1-0)^2} = \frac{1}{1} = 1$$

but $x(1-x) > 0 \therefore f(x) > 1$

When $x(1-x) = \frac{1}{4}$

$$f(x) = \frac{1 - \left(\frac{1}{4}\right)^2}{\left(1 - \frac{1}{4}\right)^2} = \frac{\frac{15}{16}}{\frac{9}{16}} = \frac{15}{9} = \frac{5}{3}$$

but $x(1-x) \leq \frac{1}{4} \therefore f(x) \leq \frac{5}{3}$

combining the inequalities

so $1 < f(x) \leq \frac{5}{3}$

Alternate solution:

$$\begin{aligned}\text{Now, } f(x) - 1 &= \frac{1+x-x^2}{1-x+x^2} - 1 = \frac{1+x-x^2-(1-x+x^2)}{1-x+x^2} = \frac{2x-2x^2}{1-x+x^2} \\ &= \frac{2x(1-x)}{\frac{3}{4} + \left(x - \frac{1}{2}\right)^2} \\ &> 0, \quad \text{since } x > 0 \text{ and } 1-x > 0 \Rightarrow x(1-x) > 0 \\ &\quad \text{and } \frac{3}{4} + \left(x - \frac{1}{2}\right)^2 \geq \frac{3}{4} > 0\end{aligned}$$

Hence $f(x) > 1$ (1)

$$\begin{aligned}\text{Also, } \frac{5}{3} - f(x) &= \frac{5}{3} - \frac{1+x-x^2}{1-x+x^2} = \frac{5(1-x+x^2) - 3(1+x-x^2)}{3(1-x+x^2)} = \frac{2-8x+8x^2}{3(1-x+x^2)} \\ &= \frac{2(1-2x)^2}{3\left(\frac{3}{4} + \left(x - \frac{1}{2}\right)^2\right)} \\ &\geq 0\end{aligned}$$

Hence $f(x) \leq \frac{5}{3}$ (2)

Combining (1) and (2) gives $1 < f(x) \leq \frac{5}{3}$.

16 c iii	$OQ = b + xBC$ $BC = -b + c$ $OQ = b + x(-b + c)$ $= (1 - x)b + xc$ and as in (i) $OQ = \sqrt{1 - x + x^2}$ $\cos \angle POQ = \frac{OP \cdot OQ}{ OP OQ }$ $= \frac{((1 - x)a + xb) \cdot ((1 - x)b + xc)}{\sqrt{1 - x + x^2} \sqrt{1 - x + x^2}}$ $= \frac{(1 - x)^2 a \cdot b + x(1 - x)a \cdot c + x(1 - x)b \cdot b + x^2 b \cdot c}{1 - x + x^2}$ $\frac{\frac{1}{2}(1 - 2x + x^2) + \frac{1}{2}(x - x^2) + x - x^2 + \frac{1}{2}x^2}{1 - x + x^2}$ $= \frac{1}{2} \frac{1 - 2x + x^2 + x - x^2 + 2x - 2x^2 + x^2}{1 - x + x^2}$ $= \frac{1}{2} \frac{1 + x - x^2}{1 - x + x^2}$ so from (ii) $\frac{1}{2} < \cos \angle POQ \leq \frac{5}{6}$ $0.585 \leq \angle POQ < \frac{\pi}{3}, \text{ or}$ $33.6^\circ \leq \angle POQ < 60^\circ$	3 marks – correct solution 2 marks – correctly expresses $\angle MON$ as a quotient of dot product and magnitude 1 mark – correct dot product fully simplified
----------------	---	--